Looking for a Needle in a Haystack: Synthetic VAT Fraud Networks for Graph Learning

[Extended abstract]

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Abstract

Access to firm-level invoices is severely restricted by tax secrecy, commercial confidentiality, and privacy requirements. Consequently, synthetic data could be an important resource for fiscal-crime research, as they have been for researchers in related areas of financial-crime detection, such as anti-money laundering. For Value Added Tax (VAT) fraud, however, we are not aware of any public benchmark designed specifically for this task. We address this gap with an agent-based framework that generates temporal invoice networks and embeds coordinated Missing Trader frauds within a realistic background economy. Firms are represented as nodes and taxable supplies as directed, weighted edges. Missing Traders create unpaid VAT liabilities, Buffers obscure the transaction path, and Brokers realise the benefit through deductions or refunds. Detection is a needle-in-a-haystack problem: tax authorities may observe millions of taxpayers and invoices, but only a few fraudsters. The resulting task is deliberately asymmetric: Missing Traders expose feature-level anomalies, whereas compliant-looking Brokers can be identified only through their multi-hop relationship to distant fraudulent sources. This combination of class imbalance, heterophily, feature camouflage, and long-range dependence makes the data useful both for practical audit research and for evaluating graph-learning models under conditions in which topology is essential rather than incidental.

Keywords

synthetic financial data, Value Added Tax fraud, agent-based modelling, invoice networks, graph learning, long-range dependencies

1. Motivation and Data Gap

Transaction-level data are indispensable for studying financial crime because many illicit activities are defined not by a single anomalous payment, but by coordinated relationships among several entities. Such records are rarely available to researchers. In the tax domain, they reveal commercially sensitive supplier–customer relationships, firm performance, and potentially personal information. Access is therefore constrained by tax secrecy, confidentiality, privacy regulation, and the operational security requirements of revenue authorities. Even where administrative data can be studied, labels generally reflect only taxpayers selected for audit and frauds already discovered. The resulting sample is delayed and affected by the non-random nature of risk-based investigations [**?**].

Synthetic transactions offer a controlled alternative. They have already been used to support research in related financial-crime settings, including anti-money-laundering model development [**?**]. However, synthetic payment datasets cannot simply be repurposed for VAT fraud: the relevant object is an invoice network, and the offence exploits the legal propagation of tax credits along a supply chain. Public graph benchmarks dedicated to this mechanism are, to the best of our knowledge, not available. Existing VAT studies instead rely on confidential administrative records that are not publicly disclosed [**? ?**]. This limits reproducibility and makes it difficult to compare methods for detecting coordinated fraud.

Scale and sparsity create a second obstacle. A national invoice network can contain millions of businesses and transactions, but a coordinated VAT scheme may be carried out by only a few dozen actors hidden among them [**? ?**]. Fraudulent entities therefore constitute an extremely small and

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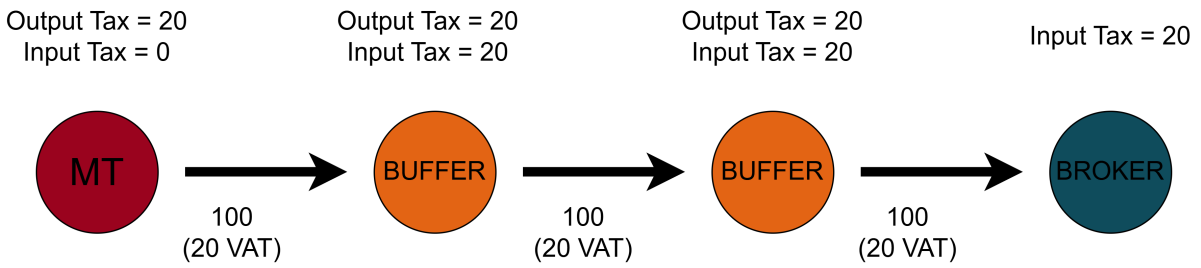


Figure 1: Simplified domestic Missing Trader fraud. The MT charges 20 units of VAT on a sale of 100 but does not remit it. Buffers relay the invoice or goods with approximately neutral VAT positions, while the Broker receives the corresponding input-tax credit. Arrows denote the direction of the taxable supply.

structured minority rather than a broadly distributed class. A useful detection method must isolate this small subgraph without overwhelming investigators with false positives from the legitimate economy.

We propose an agent-based generator of labelled VAT invoice networks. Its aim is to encode the economic, temporal, and topological mechanisms that make VAT fraud difficult. The generator provides complete ground truth for each actor and transaction. Our approach is highly configurable, allowing network size, fraud prevalence, path length, and behavioural camouflage to be varied systematically and the difficulty of the task to be controlled.

2. From VAT Collection to Missing Trader Fraud

VAT is an indirect consumption tax collected fractionally along a supply chain. Under the credit-invoice method, a registered firm charges output VAT on sales and deducts input VAT paid on purchases. Its periodic liability is

$$\text{VAT liability} = \text{output VAT} - \text{input VAT}. \tag{1}$$

For example, a firm buying goods for 100 plus 20 VAT and reselling them for 150 plus 30 VAT remits 10. The successive net payments sum to the VAT borne by the final consumer.

The same deduction mechanism creates a structural vulnerability. A Missing Trader (MT) sells goods or issues invoices with VAT and disappears or becomes insolvent before remitting it. In intra-Community fraud, the MT typically acquires goods VAT-free from another EU member state and then sells them domestically; analogous schemes can also be organised within one jurisdiction [**?** **?**]. Intermediate firms, known as Buffers, resell the goods or invoices at small margins. Some may be complicit shell companies, whereas others may be legitimate businesses. Their purpose is to lengthen and obscure the path from the MT to the beneficiary. Finally, a Broker deducts input VAT corresponding to tax that the MT never remitted. In carousel variants, the goods may be exported and circulated again through the same or a related network [**?**].

Figure **??** shows the smallest linear instance, but operational schemes are rarely isolated chains. They can contain several MTs, multiple layers of Buffers, more than one Broker, and transactions with legitimate firms [**?**]. These characteristics, together with severe class imbalance, make the detection task particularly difficult.

Given the relational nature of VAT fraud and electronic invoicing, a directed graph is the natural representation: firms are nodes, invoice or goods flows are edges, and edge attributes include time, taxable amount, and VAT. The graph direction follows the taxable supply rather than the reverse flow of money, so that an input-tax credit can be traced back towards the entity that created it.

3. Agent-Based Generation of Invoice Networks

The simulation builds the economy from the bottom up. In the current configuration, it initialises a closed population of 10,000 firms. Firm size, initial liquidity, inventory, workforce, and physical capital

are calibrated to the highly skewed composition of Italian enterprises reported by ISTAT. Legitimate trading relationships are generated with degree-sensitive selection so that the resulting invoice graph has a scale-free topology, a property observed in empirical B2B invoice networks [**?** **?**]. A segment-tree implementation supports degree-proportional counterparty sampling and updates in logarithmic time, making the generation of larger graphs feasible.

Agents interact over 365 daily time steps. Each transaction updates both parties: the seller's sales, liquidity, output VAT, and inventory; and the buyer's purchases, liquidity, inventory or physical capital, and input VAT. Outgoing invoice values include a positive, firm-specific markup over the allocated stock value. Employees and capital evolve probabilistically with economic activity rather than being independent static attributes. This coupling allows node features and graph topology to co-evolve.

Fraud is introduced by assigning a small minority of agents the roles of MT, Buffer, or Broker and organising them into groups of variable size. MTs generate high-value invoices and illicit tax credits over a deliberately short operational lifetime. Credit is routed through a variable number of Buffers, which trade at low margins, before reaching a Broker. Fraudulent entities also perform legitimate transactions, and occasional links between fraud rings prevent the labels from being recoverable through disconnected components or simple motif matching. Parameters control fraud prevalence, invoice amounts, ring size, cross-ring connectivity, and the distance between MTs and Brokers.

This mechanism differs from injecting arbitrary anomalous edges into a fixed graph. The labels correspond to economic roles, transactions modify the agents' accounts, and fraudulent subgraphs arise inside the same temporal process as ordinary commerce. Calibration against aggregate firm statistics and empirical invoice-network properties provides necessary plausibility, although synthetic fidelity must be evaluated continuously.

4. A Generated Fraud Example

Figure **??** presents fraud cluster F4 produced by the simulator and embedded in its surrounding economy. It contains four Missing Traders, twelve Buffer entities, and two Brokers, together with their links to legitimate firms. The example illustrates why a role-aware ground truth is more informative than a single binary fraud label. MTs are the sources of illicit credit; Buffers provide path latency and mix fraudulent flows with ordinary trade; Brokers are the extraction points. At the reduced scale of the simulation, F4 reproduces the same imbalance encountered in practice: a fraud scheme comprising only eighteen role-labelled entities must be recovered from a much larger legitimate network.

The roles also induce different detection regimes. An MT can exhibit a short life, little physical capital, rapid turnover growth, and an abnormal sales-to-purchases ratio. It is therefore a *feature-dominant* anomaly that may be detected by a tabular classifier. A Broker is designed to resemble a compliant firm in the same feature space. Its suspicious status depends on being the downstream beneficiary of credit originating at a distant MT. Consequently, the Broker is a *structure-dominant* anomaly: identifying it requires relational inference across the Buffer path. Legitimate-looking neighbours and different labels along that path also create a heterophilic rather than homophilic learning problem.

5. Practical and Graph-Learning Relevance

For tax administrations, detecting coordinated VAT fraud can improve risk assessment, prioritise costly audits, and reveal beneficiaries that would not be selected from their declarations alone. A simulator permits independent researchers to develop and compare more advanced models for identifying VAT fraud. In such a sparse setting, even a seemingly low false-positive rate can produce far more alerts than genuine suspects when applied to millions of entities.

For the data science community, the proposed data provide a configurable benchmark in which graph structure has a clear causal interpretation. Strong class imbalance tests methods for rare-event learning. Heterophily challenges neighbourhood-aggregation assumptions. Role labels enable evaluation beyond aggregate accuracy, which can hide total failure on Brokers. Most importantly, feature camouflage

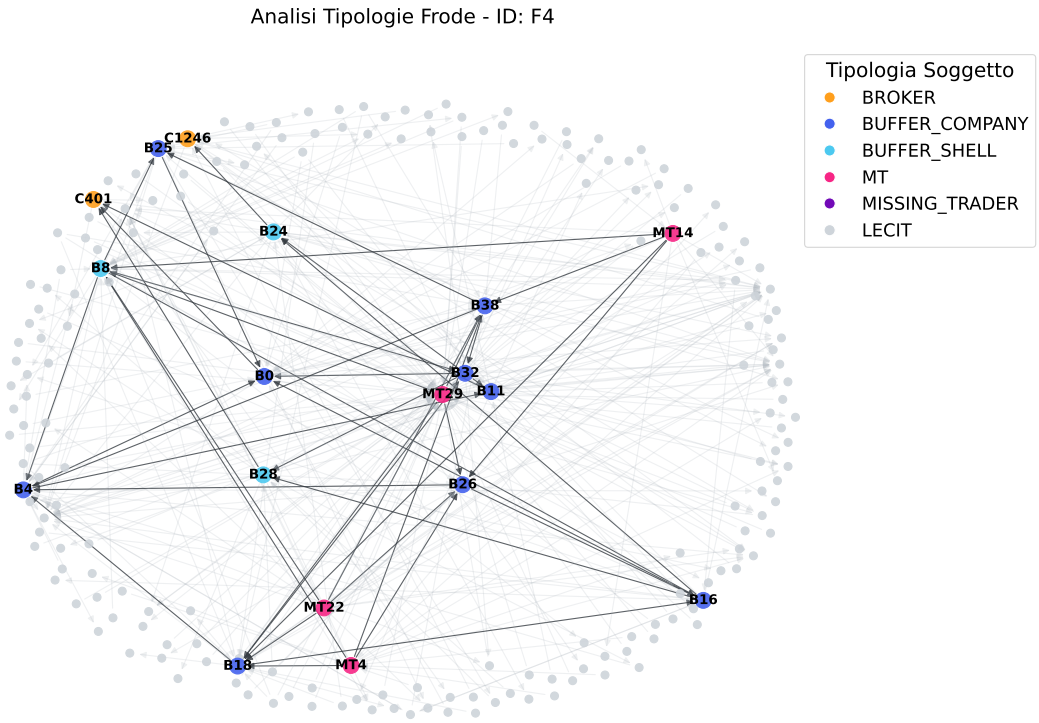


Figure 2: Synthetic fraud cluster F4 in its legitimate network context. Colours distinguish Brokers, company and shell Buffers, Missing Traders, and legitimate firms. The surrounding legitimate transactions camouflage a multi-actor, multi-hop fraudulent structure.

makes the learning task hierarchical: a model must first recognise a distant, feature-visible MT and then propagate that evidence through several legitimate-looking Buffers to classify a Broker. Increasing this distance creates a controlled long-range dependency. Standard message-passing Graph Neural Networks can lose such information through over-squashing, as signals from expanding neighbourhoods are compressed into fixed-size representations [**?**].

6. Conclusion

Privacy and confidentiality make public VAT transaction data exceptionally scarce. The offence itself is relational and difficult to represent with ordinary tabular benchmarks. By simulating the credit-invoice mechanism and embedding role-structured fraud in a temporal B2B network, the proposed framework connects a practical audit problem with open questions in graph learning. It offers reproducible ground truth for studying coordinated fraud, feature camouflage, heterophily, and long-range inference, while retaining the controls needed to test where current methods fail.

Declaration on Generative AI

During the preparation of this work, the authors used Cursor (GPT-5.6 Sol) for text drafting and language editing. The authors reviewed and edited the content and take full responsibility for the publication's content.